

Math 1314 Lesson 05

1) Solve the equation $3x^2 - 7x + 12 = 0$ by using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

a) $a = \underline{\quad ? \quad}$ $b = \underline{\quad ? \quad}$ $c = \underline{\quad ? \quad}$

b) $\sqrt{(b)^2 - 4(a)(c)} = \underline{\quad ? \quad}$

c) $x = \underline{\quad ? \quad}$

2) Solve the equation $2x^2 - 5x = 12$ by using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Hint: Add -12 to both sides first.

a) $2x^2 - 5x + \underline{\quad? \quad} = 0$

b) $a = \underline{\quad? \quad}$ $b = \underline{\quad? \quad}$ $c = \underline{\quad? \quad}$

c) $\sqrt{(b)^2 - 4(a)(c)} = \underline{\quad? \quad}$

d) $x = \underline{\quad? \quad}$

3) Solve the equation $4x^2 = -3x + 11$ by using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Hint: Add $3x$ to both sides first; then add -11 to both sides.

a) $4x^2 + \underline{\quad? \quad} + \underline{\quad? \quad} = 0$

b) $a = \underline{\quad? \quad}$ $b = \underline{\quad? \quad}$ $c = \underline{\quad? \quad}$

c) $\sqrt{(b)^2 - 4(a)(c)} = \underline{\quad? \quad}$

d) $x = \underline{\quad? \quad}$

4) Solve the equation $0.2x^2 - 0.11x = 5.2$ by using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Hint: Add -5.2 to both sides first.

a) $0.2x^2 - 0.11x + \underline{\quad? \quad} = 0$

b) $a = \underline{\quad? \quad}$ $b = \underline{\quad? \quad}$ $c = \underline{\quad? \quad}$

c) $\sqrt{(b)^2 - 4(a)(c)} = \underline{\quad? \quad}$

d) $x = \underline{\quad? \quad}$

